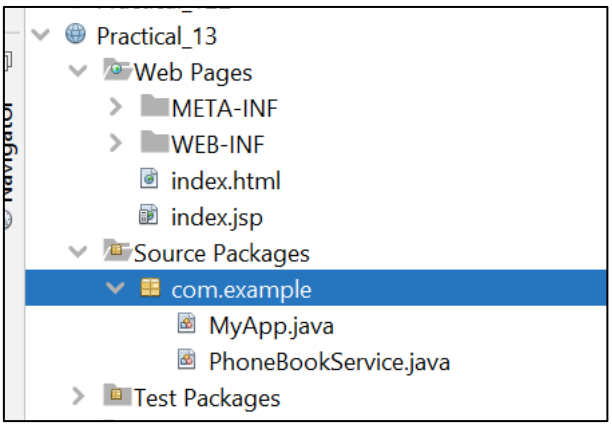
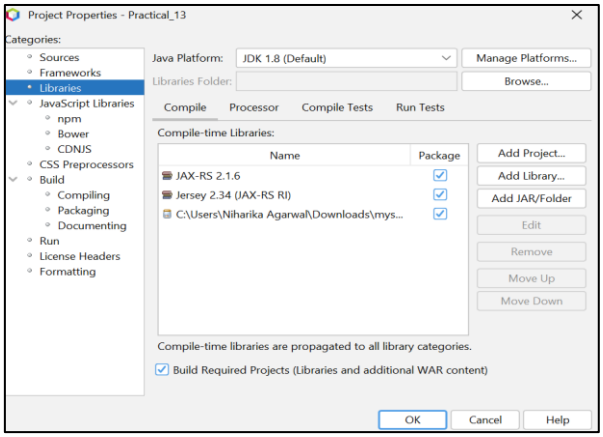
PRACTICAL 13

RESTful Web Service - CRUD Operations Develop a JAX-RS service that provides CRUD database operation interface for a phonebook. Call appropriate service from servlet/jsp pages to perform data oriented operations.





CODE:

index.jsp

```
<h2>Phonebook CRUD</h2>


<form action="api/phone/add" method="get">

    Name: <input type="text" name="name">
```

```
    <input type="submit" value="Add">
</form>

<br>

<form action="api/phone/update" method="get">
    ID: <input type="text" name="id">
    Name: <input type="text" name="name">
    <input type="submit" value="Update">
</form>
<br>
<form action="api/phone/delete" method="get">
    ID: <input type="text" name="id">
    <input type="submit" value="Delete">
</form>

<br>

<a href="api/phone/view">View All</a>
```

PhoneBookService.java

```
package com.example;

import javax.ws.rs.*;
import java.sql.*;
```

```
@Path("/phone")
public class PhoneBookService {

    private Connection con;

    // Constructor (runs once)
    public PhoneBookService() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/phonebook",
                "root", "");
        } catch (Exception e) {
            System.out.println(e);
        }
    }

    // CREATE
    @GET
    @Path("/add")
    public String add(@QueryParam("name") String name) {
        try {
            PreparedStatement ps = con.prepareStatement(
                "INSERT INTO contacts(name) VALUES(?)");
            ps.setString(1, name);
            ps.executeUpdate();
```

```
        return "Added";
    } catch (Exception e) {
        return e.toString();
    }
}

// READ
@GET
@Path("/view")
public String view() {
    String r = "";
    try {
        ResultSet rs = con.createStatement()
                .executeQuery("SELECT * FROM contacts");

        while (rs.next())
            r += rs.getInt(1) + " : " + rs.getString(2) + "\n";

    } catch (Exception e) {
        r = e.toString();
    }
    return r;
}

// UPDATE
@GET
```

```
@Path("/update")
public String update(@QueryParam("id") int id,
                     @QueryParam("name") String name) {
    try {
        PreparedStatement ps = con.prepareStatement(
            "UPDATE contacts SET name=? WHERE id=?");
        ps.setString(1, name);
        ps.setInt(2, id);
        ps.executeUpdate();
        return "Updated";
    } catch (Exception e) {
        return e.toString();
    }
}

// DELETE
@GET
@Path("/delete")
public String delete(@QueryParam("id") int id) {
    try {
        PreparedStatement ps = con.prepareStatement(
            "DELETE FROM contacts WHERE id=?");
        ps.setInt(1, id);
        ps.executeUpdate();
        return "Deleted";
    } catch (Exception e) {
```

```
            return e.toString();
        }
    }
}
```

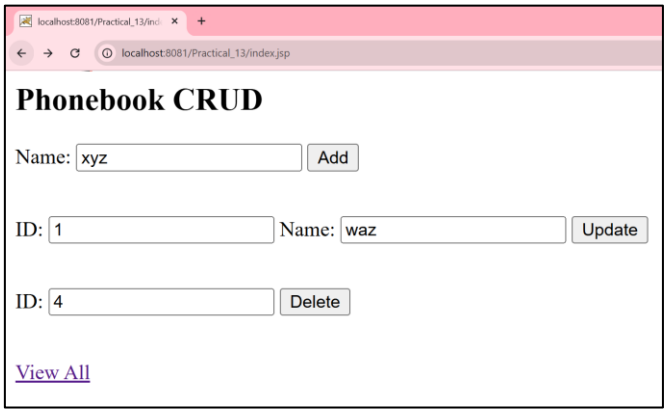
MyApp.java

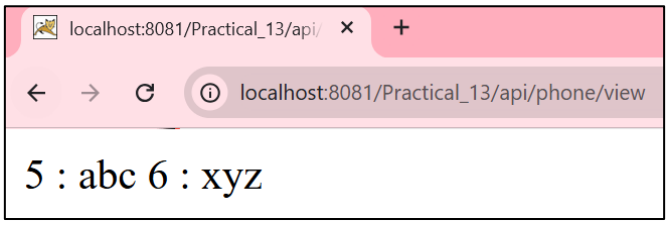
```
package com.example;

import javax.ws.rs.ApplicationPath;
import javax.ws.rs.core.Application;

@ApplicationPath("/api")
public class MyApp extends Application {
}
```

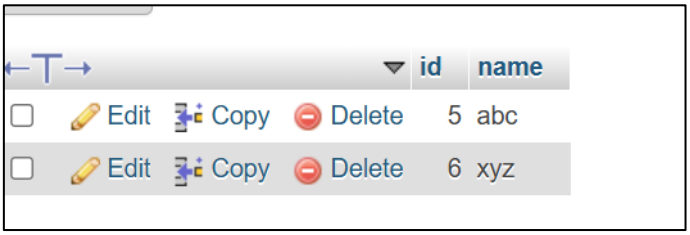
Database:

```
Create database phonebook;
CREATE TABLE contacts (
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100)
);
```

OUTPUT:


localhost:8081/Practical_13/index.jsp
Phonebook CRUD
Name: xyz
Add
ID: 1
Name: waz
Update
ID: 4
Delete
View All


localhost:8081/Practical_13/api/
localhost:8081/Practical_13/api/phone/view
5 : abc 6 : xyz


id
name
Edit
Copy
Delete
5
abc
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6
xyz